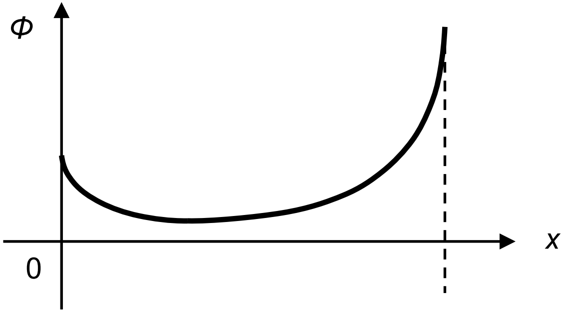
3	(a)	Similarity (any one) • Both forces acts at a distance OR both forces do not require contact (are non-contact forces). • Both forces are stronger when the two masses or charges (accept: objects, etc) are closer to each other (and weaker further apart.) OR both forces are inversely proportional to the square of the distance between the two objects. • Both are conservative forces	
		Difference (any one) • Electric force only acts on charges while gravitational force acts on masses • Electric force can be attractive or repulsive while gravitational force is always attractive.	
	(b)	(i)	charge of $\alpha = 2\left(1.6\times10^{-19}\right) \quad = 3.2\times10^{-19}$ C
			charge of $Au = 79\left(1.6\times10^{-19}\right) \quad = 1.26\times10^{-17}$ C
		(ii)	$20\text{ keV} = \left(20\times10^{3}\right)\left(1.6\times10^{-19}\right)$ $\left(= 3.2\times10^{-15}\text{ J}\right)$
			Total inital energy = Total final energy $\left(3.2\times10^{-15}\right) + 0 = 0 + \dfrac{1}{4\pi\left(8.85\times10^{-12}\right)}\dfrac{\left(1.26\times10^{-17}\right)\left(3.2\times10^{-19}\right)}{d_{\text{min}}}$
			$d_{\text{min}} = 1.13\times10^{-11}$ m

(iii)



Positive U-shaped graph, must give a finite potential value at the surface

Min point nearer to alpha, Φ larger near gold